Dr Benjamin Lee Davis
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Email: benjaminleedavis007@gmail.com

Work Experience:

New York University Abu Dhabi
509 1/2 S Main Street
Galena, KS 66739-1219 United States

09/2020 - Present
Salary: 60,000.00 USD Per Year
Hours per week: 40
Research Associate
Duties, Accomplishments and Related Skills:
Center for Astro, Particle, and Planetary Physics (CAP3) Fellow
Supervisor: Andrea Maccio (+97126284386)
Okay to contact this Supervisor: Yes

Swinburne University of Technology
Mail Number H20, PO Box 218
Hawthorn, Victoria Australia

08/2016 - 07/2020
Salary: 93,896.00 USD Per Year
Hours per week: 40
Postdoctoral Fellow
Duties, Accomplishments and Related Skills:
Reducing and analyzing galaxy images via bulge/bar/disc decompositions. I will be responsible for modeling the radial distribution of optical and/or near-infrared stellar light in nearby (less than 200 Mpc) galaxies with an emphasis on discoveries and publications related to the (central black hole)-(host bulge) connection and/or compact massive spheroids and/or galaxies with partially depleted cores or additional nuclear components.
Supervisor: Alister Graham (+61392148784)
Okay to contact this Supervisor: Yes

Arkansas Tech University
1701 N. Boulder Avenue
Russellville, AR 72801-2222 United States

08/2015 - 05/2016
Salary: 49,500.00 USD Per Year
Hours per week: 40
Visiting Assistant Professor of Physics
Duties, Accomplishments and Related Skills:
I taught Introduction to Physical Sciences, Introduction to Physical Sciences Laboratory, Physics I Laboratory, Physics II Laboratory, and General Physics II.
Supervisor: Jim Musser (+14799680361)
Okay to contact this Supervisor: Yes

University of Arkansas
825 West Dickson Street
Fayetteville, AR 72701 United States

08/2015 - 12/2015
Salary: 9,000.00 USD Piece Work
Hours per week: 20
Visiting Assistant Professor of Physics
Duties, Accomplishments and Related Skills:
Taught University Physics I and supervised its laboratories.
Supervisor: Julio Gea-Banacloche (+14795757240)
Okay to contact this Supervisor: Yes

Education:

University of Arkansas Fayetteville, AR United States
Doctorate 5 /2015
GPA: 4.0000 of a maximum 4.0000
Credits Earned: 117 Semester Hours
Major: Space and Planetary Sciences
Relevant Coursework, Licenses and Certifications:
Visiting Student Research Program Intern at the Jet Propulsion Laboratory (2011).

Pittsburg State University Pittsburg, KS United States
Bachelor's Degree 5 /2008
GPA: 3.9397 of a maximum 4.0000
Credits Earned: 199 Semester Hours
Major: Mathematics and Physics **Minor:** Music **Honors:** Magna Cum Laude

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Job Related Training:

Visiting Student Research Program Intern at Jet Propulsion Laboratory (2011)

Language Skills:

Language	Spoken	Written	Read
English	Advanced	Advanced	Advanced

Affiliations:

ARC Centre of Excellence for Gravitational Wave Discovery - Affiliate
LISA Consortium - Member

Professional Publications:

18) Sahu, N., Graham, A. W., & Davis, B. L., “Defining the (Black Hole)–Spheroid Connection with the Discovery of Morphologydependent Substructure in the MBH–nsph and MBH–Re,sph Diagrams: New Tests for Advanced Theories and Realistic Simulations,” The Astrophysical Journal, 903(2), 97, November 6, 2020.

17) Abdeen, S., Kennefick, D., Kennefick, J., Miller, R., Shields, D. W., Monson, E., & Davis, B. L., “Determining the CoRotation Radii of Spiral Galaxies Using Spiral Arm Pitch Angle Measurements at Multiple Wavelengths,” Monthly Notices of the Royal Astronomical Society, 496, 1610, June 8, 2020.

16) Sahu, N., Graham, A. W., & Davis, B. L., “Revealing Hidden Substructures in the MBH–s Diagram, and Refining the Bend in the L–s Relation,” The Astrophysical Journal, 887, 10, December 10, 2019.

15) Davis, B. L., Graham, A. W., & Combes, F., “A Consistent Set of Empirical Scaling Relations for Spiral Galaxies: The (vmax, MDM)–(s0, MBH, f) Relations,” The Astrophysical Journal, 877, 64, May 24, 2019.

14) Sahu, N., Graham, A. W., & Davis, B. L., “Black Hole Mass Scaling Relations for EarlyType Galaxies. I. MBH–M*,sph and MBH–M*,gal,” The Astrophysical Journal, 876, 155, May 15, 2019.

13) Miller, R., Kennefick, D., Kennefick, J., Shameer Abdeen, M, Monson, E., Eufrasio, R. T., Shields, D. W., & Davis, B. L., "Investigating the Origins of Spiral Structure in Disk Galaxies Through a Multiwavelength Study," The Astrophysical Journal, 874, 177, April 5, 2019.

12) Graham, A. W., Soria, R., & Davis, B. L., "Expected intermediate mass black holes in the Virgo cluster. II. Latetype galaxies," Monthly Notices of the Royal Astronomical Society, 484, 814, March 21, 2019.

11) Davis, B. L., Graham, A. W., & Cameron, E., "Black Hole Mass Scaling Relations for Spiral Galaxies. I. MBH–M*,sph," The Astrophysical Journal, 873, 85, March 6, 2019.

10) Davis, B. L., Graham, A. W., & Cameron, E., "Black Hole Mass Scaling Relations for Spiral Galaxies. II. MBH–M*,tot and MBH–M*,disk," The Astrophysical Journal, 869, 113, December 17, 2018.

9) Davis, B. L., Graham, A. W., & Seigar, M. S., "Updating the (Supermassive Black Hole Mass) – (Spiral Arm Pitch Angle) Relation: A Strong Correlation for Galaxies with Pseudobulges," Monthly Notices of the Royal Astronomical Society, 471, 2187, October 21, 2017.

8) Koliopanos, F., Ciambur, B. C., Graham, A. W., Webb, N. A., Coriat, M., MutluPakdil, B., Davis, B. L., Godet, O., Barret, D., & Seigar, M. S., "Searching for Intermediate Mass Black Holes in Dwarf Galaxies with Low Luminosity AGN: A Multiplemethod Approach," Astronomy & Astrophysics, Volume 601, A20, Dec 2016.

7) MutluPakdil, B., Seigar, M. S., & Davis, B. L., "The Local Black Hole Mass Function Derived from the MBH - P and the MBH - n Relations," The Astrophysical Journal, 830, 117, Oct 2016.

6) PourImani, H., Kennefick, D., Kennefick, J., Davis, B. L., Shields, D. W., & Shameer Abdeen, M., "Strong Evidence for the Densitywave Theory of Spiral Structure in Disk Galaxies," in The Astrophysical Journal Letters, 827, L2, Aug 2016.

5) Davis, B. L., Kennefick, D., Kennefick, J., Westfall, K. B., Shields, D. W., Flatman, R., Hartley, M. T., Berrier, J. C., Martinsson, T. P. K., & Swaters, R. A., "A Fundamental Plane of Spiral Structure in Disk Galaxies," The Astrophysical Journal Letters, 802, L13, Mar 2015.

4) Seigar, M.S., Davis, B.L., Berrier, J.C., & Kennefick, D., "Constraining Dark Matter Halo Profiles and Galaxy Formation Models Using Spiral Arm Morphology. II. Dark and Stellar Mass Concentrations from 13 Nearby FaceOn Galaxies," The Astrophysical Journal, 795, 90, Nov 2014.

3) Davis, B. L., Berrier, J. C., Johns, L., Shields, D. W., Hartley, M. T., Kennefick, D., Kennefick, J., Seigar, M. S., & Lacy, C. H. S., "The Black Hole Mass Function Derived from Local Spiral Galaxies," The Astrophysical Journal, 789, 124, Jul 2014.

2) Berrier, J. C., Davis, B. L., Kennefick, D. Kennefick, J., Seigar, M. S., Barrows, R. S., Hartley, M. T., Shields, D. W., Bentz, M. C., & Lacy, C. H. S., "Further Evidence for a Supermassive Black Hole – Pitch Angle Relation," The Astrophysical Journal, 769, 132, Jun 2013.

1) Davis, B. L., Berrier, J. C., Shields, D. W., Kennefick, J., Kennefick, D., Seigar, M. S., Lacy, C. H. S., & Puerari I., "Measurement of Galactic Logarithmic Spiral Arm Pitch Angle Using TwoDimensional Fast Fourier Transform Decomposition," The Astrophysical Journal Supplement Series, 199, 33, Apr 2012.

References:

Name	Employer	Title	Phone	Email
Andrea Maccio (*)	New York University Abu Dhabi	Associate Professor and Program Head of Physics	+97126284386	maccio@nyu.edu
Alister Graham (*)	Swinburne University of Technology	Professor	+61392148784	agraham@astro.swin.edu.au
Julia Kennefick (*)	University of Arkansas	Associate Professor	+14795755916	jkennef@uark.edu

Name	Employer	Title	Phone	Email
Daniel Kennefick (*)	University of Arkansas	Associate Professor	+14795756784	danielk@uark.edu
Marc Seigar (*)	University of Minnesota Duluth	Associate Dean	+12187266704	msseigar@d.umn.edu
Joel Berrier (*)	University of Nebraska at Kearney	Assistant Professor	+13088658282	berrierjc@unk.edu
Robert Mueller (*)	University of Arkansas	Professor	+14795755879	mueller@uark.edu
Julio Gea-Banacloche (*)	University of Arkansas	Professor and Department Chair	+14795757240	jgeabana@uark.edu
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(*) Indicates professional reference

Additional Information:

Typing Proficiency: 67 WPM (285 CPM) with 100% accuracy

LaTeX, MATLAB, Mathematica, R, Adobe Creative Cloud, IRAF, DS9, Gimp, Supermongo, Microsoft Office Suite, Unix/Linux, Fortran, C++, IDL, HTML, Google Analytics, Python, MPI, SQL, HPC, Problem Solving, Data Science, Data Engineering, Machine Learning, Data Analysis, Observing, Classical Music, Music Performance, Oration, Teaching, Logarithmic Spiral Arm Pitch Angle, Multicomponent Galaxy Decomposition, Telescopes, Black Holes, Data Reduction, Galaxy Evolution, Astronomy, Astrophysics, Active Galactic Nuclei, Galactic Structure, Galaxy Dynamics, Quasars, Galaxy Clusters, Dark Matter, Galaxy Formation, Planetary Science, Martian Surface Chemistry, Golf, Tennis, Baseball, Basketball, Violin, Viola,
Guitar, Bass, Piano, Ukulele, Saxophone, Singing, SCUBA, Carpentry, Woodworking, etc.